

Question

A pure sample of a certain inorganic salt **X** is placed in air and heated to 220°C . The thermogravimetric analysis curve shows that the sample begins to lose weight. When 66.7% (mole fraction) of **X** has reacted, the temperature is rapidly increased to 240°C , and the remaining substances interact, and the sample loses weight according to another reaction. When heated to constant weight, there is only one inorganic salt in the sample. Continuing to increase the temperature to 315°C , the sample enters a new weight loss platform. At the end of the platform, the final sample mass is 86.76% of the initial mass.

The following information is available:

1. It is known that during the weight loss process, the reduced mass is only removed in the form of H_2O molecules.
2. After the 315°C weight loss platform ends, the residual solid does not contain the element H.
3. **X** contains only one metal element.

Based on the above information, select the correct statement(s):

1. **X** is composed of three elements.
2. Four substances are involved in the entire reaction process.
3. After balancing and simplifying the reaction equation for the first heating stage, the coefficients on both sides are the same.
4. The numerical ratio of the metal element to the oxygen element in the chemical formula of the inorganic salt product produced in the second heating stage is 3 : 10.
5. The ratio of the mass of water lost in the second heating stage to the third heating stage is 2 : 1.

A. 1

B. 1, 4

C. 1, 5

D. 2, 3

E. 2, 4

F. 3, 4

G. 2, 3, 5

H. 3, 4, 5

I. 4, 5

J. 4

K. 5

L. All of the above options are incorrect.

Answer

Correct Answer: **F**

Detailed Explanation

Let the chemical formula molar mass of **X** be $M(X)$ g/mol, and the initial amount of substance be n mol. The final product is represented as **Y**.

Initial mass: $m(X) = n \times M(X)$.

Final mass: $m(Z) = 0.8676 \times m(X) = 0.8676 \times n \times M(X)$.

Total mass loss (H_2O): $\Delta m = (1 - 0.8676) \times n \times M(X) = 0.1324 \times n \times M(X)$

According to the law of conservation of mass, the mass loss is equal to the mass of water removed. Let each molecule of **X** ultimately lose k molecules of H_2O .

$$\Delta m = n \times k \times M(H_2O) = n \times k \times 18$$

Establish the equation: $n \times k \times 18 = 0.1324 \times n \times M(X)$

Obtain:

$$M(X) = (18 \times k) / 0.1324 \approx 135.95 \times k$$

CHECKPOINT

1 PTS

$$M(X) = (18 \times k) / 0.1324 \approx 135.95 \times k$$

If $k = 1$, $M(X) \approx 135.95 \times 1 = 136.0$ g/mol;

If $k = 2$, $M(X) \approx 135.95 \times 2 = 271.9 \text{ g/mol}$;

If $k = 3$, $M(X) \approx 135.95 \times 3 = 407.9 \text{ g/mol}$

First try the case of $k = 1$. At this time, $M(X) \approx 136.0 \text{ g/mol}$, each molecule of **X** ultimately loses 1 molecule of H_2O . It is known that the final product **Y** does not contain the element H, it can be known that the molecular formula of **X** contains 2 H atoms, and the chemical formula molar mass of **Y** is $M(Y) = M(X) - 18 = 118.0 \text{ g/mol}$.

CHECKPOINT

1 PTS

When $k = 1$, the chemical formula molar mass of **X** is $M(X) = 136.0 \text{ g/mol}$

CHECKPOINT

1 PTS

When $k = 1$, the chemical formula molar mass of **Y** is $M(Y) = 118.0 \text{ g/mol}$

It is known that **X** contains only one metal element and has 2 H atoms, the molar mass of the remaining acid radical is $136.0 - 2 - M(M) \text{ g/mol}$;

Common metals include alkali metals, alkaline earth metals, etc.; According to the above information, we can exhaustively try:

When $M = \text{K}$, the acid radical is PO_4^{3-} ; The molar mass is 93 g/mol . **X** is KH_2PO_4 ; and **Y** is KPO_3 ; $M(Y) = 118.0 \text{ g/mol}$.

The inorganic salt in the question undergoes 3 mass loss platforms, indicating that H_2O is lost in all three processes; Let the inorganic salt generated on the first platform be **P1**; the inorganic salt generated on the second platform be **P2**; the final product generated on the third platform is **Y**.

According to the question, the first platform consumes $\frac{2}{3}n$ **X** to generate **P1**, and on the second platform $\frac{1}{3}n$ **X** just reacts completely with **P1** to generate **P2**. It can be known that $\frac{1}{3}n$ **P1** is generated on the first platform, and $\frac{1}{3}n$ **P2** is generated on the second platform. That is, the coefficient ratio of the first reaction equation: **X** : **P1** = 2 : 1; the coefficient ratio of the second reaction equation: **X** : **P1** : **P2** = 1 : 1 : 1.

CHECKPOINT

1 PTS

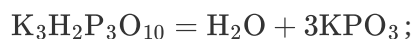
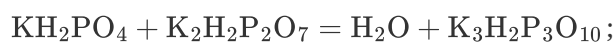
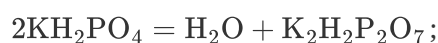
The coefficient ratio of the first reaction equation: **X** : **P1** = 2 : 1

CHECKPOINT

1 PTS

The coefficient ratio of the second reaction equation: **X** : **P1** : **P2** = 1 : 1 : 1

According to the above information, try to write the mass loss equation of KH_2PO_4 :



The overall equation is:



CHECKPOINT

2 PTS

The first reaction equation: $2\text{KH}_2\text{PO}_4 = \text{H}_2\text{O} + \text{K}_2\text{H}_2\text{P}_2\text{O}_7$

CHECKPOINT

2 PTS

The second reaction equation: $\text{KH}_2\text{PO}_4 + \text{K}_2\text{H}_2\text{P}_2\text{O}_7 = \text{H}_2\text{O} + \text{K}_3\text{H}_2\text{P}_3\text{O}_{10}$

CHECKPOINT

2 PTS

The third reaction equation: $\text{K}_3\text{H}_2\text{P}_3\text{O}_{10} = \text{H}_2\text{O} + 3\text{KPO}_3$

CHECKPOINT

2 PTS

The overall reaction equation: $\text{KH}_2\text{PO}_4 = \text{H}_2\text{O} + \text{KPO}_3$

This process conforms to the question. Therefore, **X** is KH_2PO_4 ; **Y** is KPO_3 .

CHECKPOINT

2 PTS

X is KH_2PO_4

CHECKPOINT

2 PTS

Y is KPO_3

Therefore, statements 3 and 4 are correct, choose option F.